

Student-Faculty Co-Creation of Open Educational Resources for Learning Applied Statistics with Open Source Software Tools

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Abstract -

In most university courses, students learn from textbooks they did not help develop. We present an innovative approach where undergraduate students collaborated with faculty to create an open-access, interactive web-based e-book for an introductory applied statistics course. This process transforms undergraduates into co-authors rather than passive readers, fosters an appreciation for reproducible research, encourages academic collaboration, and offers hands-on opportunities to acquire and apply new skills and technologies. Faculty also benefit from fresh ideas and new perspectives. This type of collaboration is important to explore since evidence on student–faculty partnerships on developing course materials is sparse in the literature, particularly in a Canadian context and in the area of statistics. In this paper, we illustrate the contributions of undergraduate students to curricular innovation in the development of the e-book by describing the development process, methodology and software tools used, and reflect on our experience in the process of being collaborators alongside faculty. Our participation deepened our own learning while producing resources to benefit future learners, we highlight the potential of open-source technologies and student–faculty collaboration to support the development of inclusive, adaptive, and accessible resources for statistical education.

Keywords: Bookdown, statistics education, curriculum development, interactive technology

1 INTRODUCTION

Undergraduate courses designed through a partnership between faculty and students can deepen engagement and develop a meaningful learning experience [2], [12]. Students are provided with the opportunity to be co-creators, rather than simply passive recipients of information, thereby encouraging them to take ownership of their educational journey, and faculty gain fresh insights into how students learn [12]. These partnerships between deepen engagement and promote more authentic learning experiences for both students and faculty involved [1], [12]. Systematic reviews highlight that such collaborations enhance motivation, empower students, and enrich teaching practice, yet they have not been studied or documented uniformly across disciplines [2], [16] and reports of whole-course co-creation are particularly limited within statistics education [13]. Systematic reviews by Mercer et.al [16] and Omland et. al discuss several studies where students and faculty have partnered in course designed internationally, however there is a lack of such collaboration documents in relation to mathematics and statistics education [7]. To the best of our knowledge, we could not find any collaborations specifically in a Canadian context.

Furthermore, many undergraduate statistics courses introduce foundational concepts through traditional lectures and texts. Even digital versions of textbooks are still usually static with limited advantages beyond quickly searching for content and conveniently applying bookmarks. While these approaches build necessary theoretical understanding, they are often unable to provide opportunities for students to interact meaningfully with statistical methods or explore real-world applications. Topics such as hypothesis testing, confidence intervals, and regression frequently may remain abstract due to limited context and passive delivery formats **[PLEASE FIND CITATION TO JUSTIFY THIS]**.

Our project emerged in response to those to the lack of study in student-faculty collaboration in co-developing course material and to simultaneously address the limitations of traditional texts in statistics. We partnered with faculty through the *STA378: Independent Research Project Course*, at the University of Toronto Mississauga (UTM) to assist the current course coordinator, for *STA258: Statistics with Applied Probability*, develop an interactive e-book for the course as part of an on-going course redevelopment project. STA258 is an introductory applied statistics course at UTM where most students learn about applied inference techniques (i.e. testing of hypotheses, interval estimation) and regression for the first time in their university learning experience. This is an important course since it provides students with the foundational skills for data analysis and equips them with the ability to interpret statistical results in advanced courses across all disciplines, including other statistics courses, the life sciences, social sciences, economics and finance and computer science courses requiring a knowledge of inference. This course also provides students with the ability to read and interpret statistical results from academic journals which is essential for critically evaluating research, making evidence-based decisions, and succeeding in advanced study and professional practice in both academic and non-academic settings.

The current course coordinator is involved in a long term project to redesign the content in STA258. There are several reasons justifying the development of STA258. One such reason is that it currently overlaps considerably with *STA260: Probability and Statistics II*, with both courses sharing common introductory content. Students simultaneously enrolled in both courses experience substantial duplication in the content delivered to them during the opening weeks of both courses. Another reason to redevelop STA258 is the necessity to incorporate more data analysis components to modernize the course and make it more relevant for students by equipping them with foundational skills for independent statistical analysis. There are some distinct features of STA258 which makes it a good candidate for collaborative re-design between students and faculty. Within the University of Toronto system, STA258 is only offered at the Mississauga campus and there is no equivalent course or combination of courses available at the St. George campus or the Scarborough campus. Therefore, it has the potential to be a self-contained course which can equip students with the skills required providing an efficient pathway for students to learn important skills related to statistical reasoning, which is equivalent to multiple courses from other campuses. Redesigning STA258 eliminates the necessity for the Department of Mathematical and Computational Sciences at UTM to go through lengthy development and approval process required for an entirely new course to address the curricular gap.

The core team working on this project consisted of the two undergraduate authors of this paper, and two faculty members who are both Assistant Professors in Statistics at the Department of Mathematical and Computational Sciences at UTM. One of the faculty members is the current course coordinator for STA258. We contributed to content development and the technical implementation, and the faculty members provided consistent guidance through regular meetings, detailed feedback, training with software, and contributed critical input into decisions regarding the e-book's structure and tools. Both Professors also assisted with trouble shooting and the

course coordinator contributed a substantial amount towards the coding component, particularly with initiating the framework and resolving some of the more challenging components. We also received valuable feedback from two additional external partners who provided suggestions related to content scope, pedagogical structure, and overall educational value. One of the external partners is faculty at a College in British Columbia and the other is an expert consultant in communication and design of digital assets. We met with faculty weekly for progress checks and trouble shooting and we met with our external partners approximately once a month to provide progress updates and to receive feedback. One of the authors was living outside of Canada and joined meetings through video conference.

During the initial phase of the STA258 course development timeline, we as undergraduate students worked alongside Dr. Mudalige and Dr. Ataei to produce an open-access, interactive e-book which integrates narrative exposition, reproducible simulations and examples involving code, embeds visualizations and interactive components to explain concepts, and provides an open-source resource for educators in the broader statistical community. This paper outlines the framework that guided our partnership, describes the authoring workflow (version control, peer review, and continuous deployment), and reflects on early outcomes. By sharing these first-phase results, we aim to provide a practical model for implementing student-faculty co-creation in applied statistics and related quantitative courses.

2 DESIGN, DEVELOPMENT AND IMPLEMENTATION

The interactive e-book project was launched during the Summer 2025 semester (May - August) as a joint initiative between undergraduate students and faculty to create an engaging and innovative learning resource for students enrolled in statistics courses. The University of Toronto Mississauga provides high-performing undergraduate students enrolled in a statistics major or specialist with the opportunity to enrol in *STA378: Statistics Research Project*, where students explore a topic in statistics under the supervision of a faculty member. Students can not directly enrol into STA378 through ACORN, and a special administrative process is required, including approval from the associate chair. Both authors of this paper had recently completed STA258 during the Winter 2025 semester and agreed with Dr. Mudalige that the course required adjustments, and we mutually expressed interest in working together on this course redevelopment project.

Our initial decision was to use LaTeX to create a Portable Document Format (PDF) e-book. This was based on its well-established typesetting capabilities and familiarity within academic publishing. This PDF document generate in LaTeX is also easily accessible on many devices. In addition, open-source resources, such as OpenIntro Statistics, which offer an extensive material to utilize as a foundation to develop an e-book using LaTeX. However there are limitations with this approach, including being limited to fixed sized pages to display content, limited navigation options, the difficulty with embedding interactive components, and maintaining an organized working environment due to the many auxiliary files produced.

Although we began the e-book in LaTeX, midway through our development we learned about Bookdown which is a package in R package developed for writing materials such as books and long-form content with R Markdown. Bookdown has the advantage of converting R Markdown into responsive HTML, EPUB, and print-ready PDF while automatically generating cross-references and a searchable table of contents, LaTeX can be typeset within R Markdown documents, the HTML output from Bookdown can embed live R code, interactive widgets, and interactive apps using libraries such as shiny and plotly, giving students a hands-on e-book experience that fixed-layout PDFs cannot match, and a book generated in Bookdown can be hosted

on GitHub pages allowing anyone with an internet connection to easily access it. LaTeX code written in Bookdown is rendered using MathJax which is an open-source JavaScript library for mathematical symbols and equations for web browsers. Therefore, the vast majority of content which can be typeset in LaTeX can be rendered on a browser using Bookdown. Due to its many advantages over traditional LaTeX, we decided to transition to the Bookdown framework since it proved more compatible with the goals of interactive and accessible online learning.

This shift from LaTeX to Bookdown allowed us to integrate dynamic and interactive elements that enhance student engagement and understanding and transformed static content into a fully interactive learning resource. The conversion from LaTeX (.tex) to R Markdown (.Rmd) was performed using Pandoc, with the majority of the the formatting and structure intact. Pandoc also preserved most of the fidelity of the LaTeX code during the conversion. The syntax used for the conversion is given in Listings 1 and 2.

```
1 pandoc -s input.tex -f latex -t markdown -o output.Rmd
```

Listing 1: Basic conversion of a LaTeX .tex file to an R Markdown .Rmd file using Pandoc.

```
1 pandoc --filter pandoc-citeproc --mathjax -s input.tex -f latex -t markdown -o  
output.Rmd
```

Listing 2: Conversion of a LaTeX .tex file to an R Markdown .Rmd file rendered with MathJax using Pandoc.

Once we established our framework, we selected other open-source tools and programming languages that could support both visual design and reproducible workflows around Bookdown. A summary of the technologies used in developing the e-book are listed in Table 1.

Languages	R, Markdown, LaTeX, YAML, HTML, CSS, JavaScript
Frameworks	Bookdown, R Markdown
R packages	knitr, ggplot2, plotly, Shiny, rgl
Version Control	Git (with GitHub)
Deployment	GitHub Pages

Table 1: Summary of software, libraries and frameworks used in developing the e-book.

The tools utilized in Table 1 were intentionally selected for their compatibility, transparency, and capacity to support reproducible computing. Some tools were used directly such as R Markdown and R, while others were secondary technologies which were used by the frameworks in rendering the e-book. GitHub served both as a collaborative development platform and as a public-facing repository for version control, enabled open access to the resources created and providing asynchronous contributions from others. The core members worked on developing content on various branches on GitHub before merging everything on to the main branch after sufficient progress. We utilized GitHub Pages to host the e-book directly from the repository used for the project. GitHub Pages is a service offered by GitHub which allows users to host webpages directly from a GitHub repository. The YAML file was used for front matter, to store the book's metadata, for build settings and for integration with GitHub pages CSS was used to customize environments in Bookdown, such as the definition environment and remark environment with personalized aesthetics and consistent colour scheme.

Although we encountered some technical challenges, especially during the transition from LaTeX to Bookdown, these obstacles deepened our understanding of collaborative digital authorship and reproducible research. The final product represents not only an innovative approach to content delivery, but also an exploration of what collaborative educational development can look like in practice.

The development of the e-book required significant planning and coordination to ensure that the final product was cohesive and pedagogically meaningful. The initial draft was largely a collection of lecture notes and presentation slides. We used these to build a single unified document which was a one-to-one replication of the notes and slides. This was our baseline draft for the e-book. During the development of this baseline, the inclusion of figures and graphics was sparse since we were aware that content would be rearranged. After reviewing this initial baseline draft, we realized that the large number of chapters resulted in fragmentation of related content, making it difficult for students to learn material.

Through collaboration and discussion, we combined overlapping sections and reordered material into a more logical flow. This process helped reduce redundancy and resulted in a smoother experience for both students and instructors and material being presented in a coherent manner. We also removed as much content as possible which overlapped with STA260, making these two courses much more distinct. Once the order and layout were finalized, the content created was rearranged and refined, and static and interactive plots and figures were built around the updated content.

A brief description of the specific R packages from Table 1 which were used in developing the e-book are given in Table 2:

R Package	Description
knitr	The knitr package is designed to integrate R code, output, and commentary into a single document, facilitating reproducible research and dynamic report generation.
ggplot2	A data visualization package that implements a layered grammar of graphics, enabling the systematic creation of a diverse range of static plots.
plotly	This package provides tools for generating interactive, web-based plots and dashboards that can be easily shared and embedded.
Shiny	An R package for building interactive web applications directly from R, allowing users to create powerful data analysis tools and visualizations without extensive knowledge of web development languages.
rgl	A visualization package that creates interactive 3D plots and scenes, providing a robust platform for exploring spatial data. It leverages OpenGL for high-performance rendering and can export scenes to WebGL, allowing for the display of 3D content within web browsers.

Table 2: Description of each R package used in developing the e-book.

Static and interactive plots and figures are rendered natively within the R Markdown file within code blocks. The Tikz package was initially used with LaTeX to create some static graphics which were used in the both the initial PDF and the first draft of the Bookdown version of the e-book, since Tikz is a powerful package which can create vector graphics. However, we discovered that we can create all of the required graphics using ggplot2 and plotly, so we decided to utilize these 2 packages for all figures for a more consistent appearance.

An example of a code block used to render a figure using the ggplot2 library is given in Listing 3.

```
1  ```{r fig_label, fig.cap="Example_Code_Block", echo=FALSE, fig.align='center',  
    warning=FALSE}  
2  
3  library(ggplot2)  
4  # Code for ggplot2 figure goes here  
5  # The ggplot2 figure is natively created and natively embedded into HTML when  
    rendered in Bookdown  
6  ```
```

Listing 3: An example of a code block in Markdown used to render an image with ggplot.

Interactive plots were created using the Plotly and R Shiny. Plotly figures are generated natively within codeblocks similar in a similar manner to ggplot2, with syntax similar to that of 3. R Shiny apps were developed in R and hosted on Shinyapps.io, which is a platform provided for hosting and sharing Shiny web applications. Shinyapps.io is maintained by Posit, who are most well known as the developers of R Studio and Positron. These apps are natively embedded into the e-book using a code block and some HTML for formatting. Listing 4 provides the syntax for embedding R Shiny apps in the e-book.

```
1  ```{r shiny_app_label, fig.cap="Example_Interactive_Application", echo=FALSE,  
    out.width='100%', warning=FALSE}  
2  
3  knitr::include_app("URL_to_App_Hosted_on_shinyapps.io")  
4  ```
```

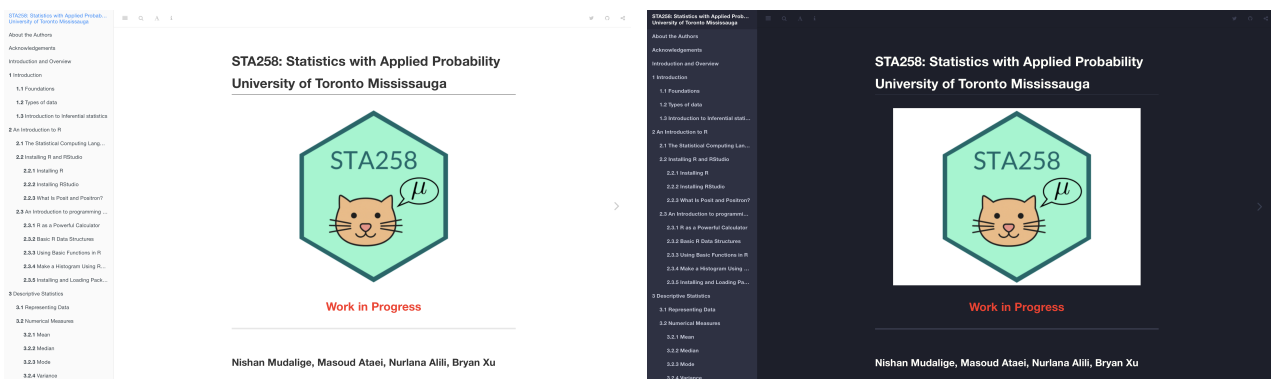
Listing 4: An example of a code block in Markdown used to render an image with ggplot

Finally, we used the rgl library to create a 3D interactive version of the digital hex sticker designed for the project which appears on the cover of the e-book. This was done for aesthetic purposes and to provide the e-book with a recognizable identity.

3 CONTENT AND INTERACTIVE DESIGN

The textbook is live and available to be accessed at nishanmudalige.github.io/STA258/. We note that the e-book is still a work in progress since it is one component which is part of an ongoing, long-term course redevelopment project which we discussed in Section 1.

We selected the default theme of Bookdown since it has a clean layout and appearance. The default theme also comes with options for light mode, dark mode and high contrast (sepia) mode with the table of contents appears along the left. The landing page in light and dark mode is presented in Figure 1. The table of contents is available on the left of the webpage and it can be hidden if desired. There is also a search feature, indicated by the magnifying glass in Figures 1a and 1b where indexing is automatically performed by Bookdown. Additional accessibility features to help improve legibility include the ability to increase or decrease font size, increase or decrease spacing between lines and the ability to switch between serif and sans serif fonts.



(a) Landing page of the e-book in the default layout.

(b) Landing page of the e-book in dark mode.

Figure 1: Landing page of the e-book in the default and dark theme. Notice the table of contents on the left.

The main chapters in the e-book are organized in the following manner:

- Chapter 1: Introduction
- Chapter 2: An Introduction to R
- Chapter 3: Descriptive Statistics
- Chapter 4: Foundations of Inference
- Chapter 5: Confidence Intervals
- Chapter 6: Hypothesis Tests
- Chapter 7: Statistical Power
- Chapter 8: Analysis of Variance
- Chapter 9: Simple Linear Regression
- Chapter 10: Analysis of Categorical Data

Chapter 1 is a broad introduction to applied statistics; Chapter 2 is a dedicated section with an introduction to R; Chapter 3 contains foundational statistical concepts, including key terminology, data types, and the basic ideas behind inferential statistics; Chapter 4 sets the stage for statistical inference covering sampling distributions and the main reference distributions in the course (the normal distribution, t -distribution, F -distribution and chi-square distribution; Chapters 5 and 6 cover standard one-sample and two-sample inference techniques on a mean, on a proportion, on a difference of means, on a difference of proportions, on paired data, on a variance and on a ratio of two variances. Chapter 7 discusses the type I and type II errors and introduces statistical power for the one-sample case when the population variance is known; Chapter 8 covers one-way analysis of variance; Chapter 9 covers simple linear regression, including inference on the slope; and Chapter 10 covers chi-square tests for categorical data. These chapters were further broken down at a more granular level into sub-sections, however the title of these components are not listed for brevity. An example of the appearance of content of the e-book is presented in Figure 2.

Recognizing the importance of technical skills, the second chapter is an introduction to R and its practical uses. This section explains how to install R, use essential functions, and work with basic data structures and visualizations. Together, these elements provide a strong technical foundation. This chapter also provides information on working with R Studio which is a popular integrated development environment (IDE) for programming in R. We also introduce positron which is the next-generation of data science IDE that also supports

3.2.1 Mean

Definition 3.1 (Mean) Let $x_1, x_2, x_3, \dots, x_n$ represent a sample of n observations. The sample mean is defined as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

3.2.2 Median

Definition 3.2 (Median) The median is the middle value of a data set when the observations are arranged in ascending order.

The calculation of the median depends on whether the sample size is odd or even. If the sample size is odd, the median is the middle observation. If the sample size is even, the median is the average of the two middle observations.

Consider n ordered observations $x_{(1)}, x_{(2)}, \dots, x_{(n)}$.

- If n is odd

$$\text{Median} = \text{observation} \left(\frac{n+1}{2} \right)$$

- If n is even,

$$\text{Median} = \text{average of observation} \left(\frac{n}{2} \right) \text{ and observation} \left(\frac{n}{2} + 1 \right)$$

(a) Sample content from the Chapter 3: Descriptive Statistics.

4.4.1 Convergence in Probability

Definition 4.9 (Convergence in Probability) The sequence of random variables $X_1, X_2, \dots$ is said to converge in probability to the constant c , if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - c| \leq \varepsilon) = 1$$

or equivalently,

$$\lim_{n \rightarrow \infty} P(|X_n - c| > \varepsilon) = 0$$

Remark. If a sequence of random variables $X_n = \{X_1, X_2, \dots\}$ converges in probability to c , we denote this as $X_n \xrightarrow{P} c$.

This concept plays a key role in the Law of Large Numbers, where the sample mean of independent and identically distributed random variables converges in probability to the population mean as the sample size grows.

Recall Chebyshev's Inequality, which provides a bound on the probability that a random variable deviates from its mean.

Theorem 4.2 (Chebyshev's Inequality) Let X be a random variable with finite mean μ and variance σ^2 . Then, for any $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Using complements:

$$P(|X - \mu| < k) \geq 1 - \frac{\sigma^2}{k^2}$$

(b) Sample content from the Chapter 4: Foundations on inference

Figure 2: Examples of written content of the e-book.

coding with python. This chapter helps the reader become familiar with R by introducing the language as a powerful calculator and also introduces some basic data structures, functions and graphical plots. This sets the necessary ground work for the coding examples which are provided in subsequent chapters.

Numerous static figures were created using `ggplot2` within code blocks in the form of Listing 3. Some examples of the static figures created are given in Figure 3.

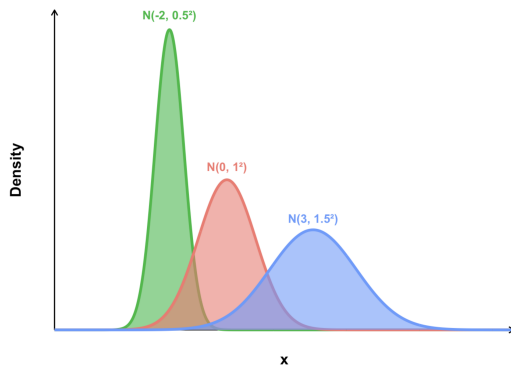


Figure 4.1: Some examples of normal distribution curves with different means and variances

(a) Multiple normal distributions created with `ggplot2` from a figure in Chapter 4: Foundations on inference.

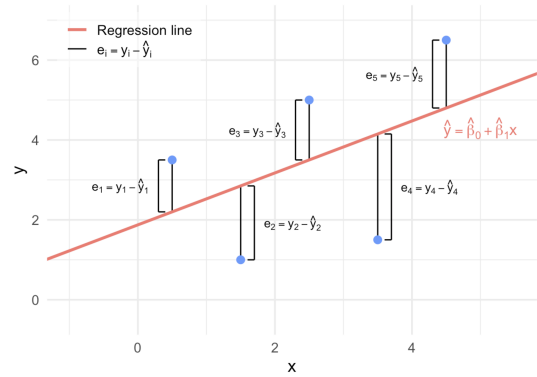


Figure 9.3: The distances between each observed y_i and fitted $\hat{y}_i$ from Figure 9.2.

(b) A figure introducing residuals created using `ggplot2` from Chapter 9: Simple Linear Regression.

Figure 3: Examples plots created using `ggplot2`.

Although `plotly` can be used for interactive figures, we also used `plotly` to create several static figures by deactivating the options which allow for interactivity. This decision was out of convenience for a more unified design of figures and for aesthetics. Not all of the figures in the e-book required an interactive component. If interactivity was forcefully implemented on to all figures, some of them would have become unnecessarily distractive for students. Examples of figures created with `plotly` in which the interactive options were removed are presented in Figure 4.

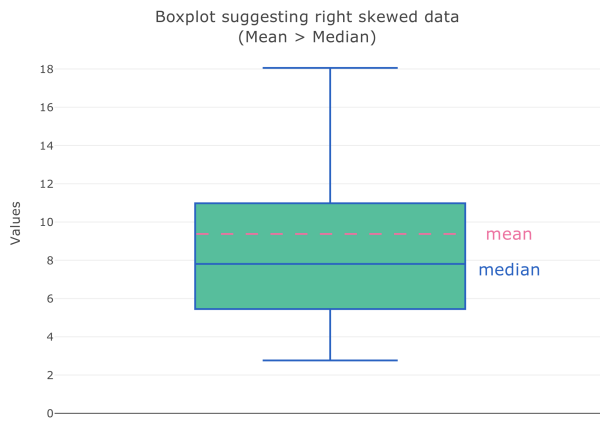


Figure 3.9: An illustration of an boxplot with right (positive) skewed.

(a) A right skewed boxplot created with `plotly` with interactive elements disabled from Chapter 3: Descriptive Statistics.

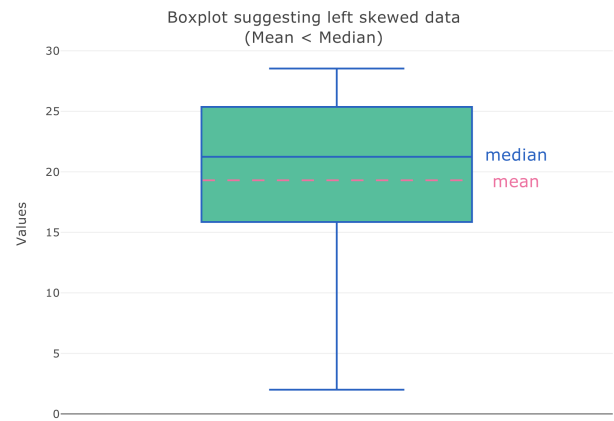


Figure 3.8: An illustration of an boxplot with left (negative) skewed.

(b) A left skewed boxplot created with `plotly` with interactive elements disabled from Chapter 3: Descriptive Statistics.

Figure 4: Examples plots created with `plotly` where the interactive options were disabled.

One of the areas in which this e-book stands out is the inclusion of interactive plots. Some interactive plots were created using `plotly` while others were created using `shiny`, where the shiny app was hosted on Shinyapps.io as discussed in Section 2. Plots with relatively simpler components were created with `plotly`, such as plots which would display or hide information when the user hovers over them, and plots which had buttons on them to show or hide a feature. An example of such a plot created with `plotly` is illustrated in Figure 5.

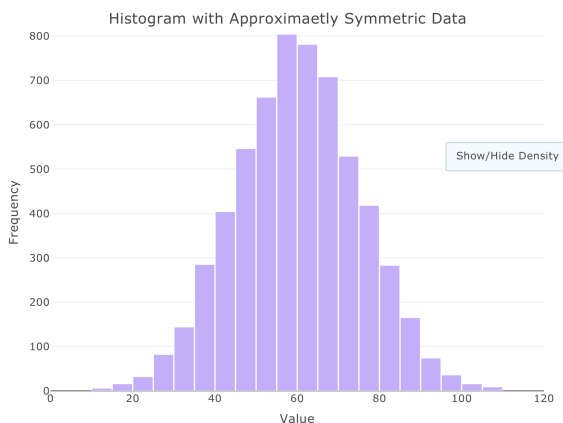


Figure 3.6: An illustration of a histogram to have a symmetric probability distribution.

(a) Interactive histogram created with `plotly` with density plot hidden from Section 3: Descriptive Statistics.

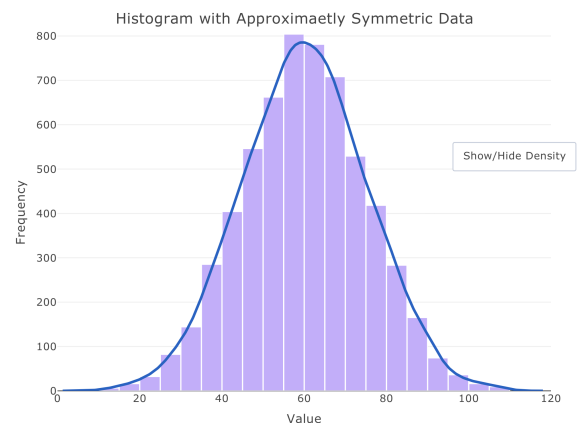


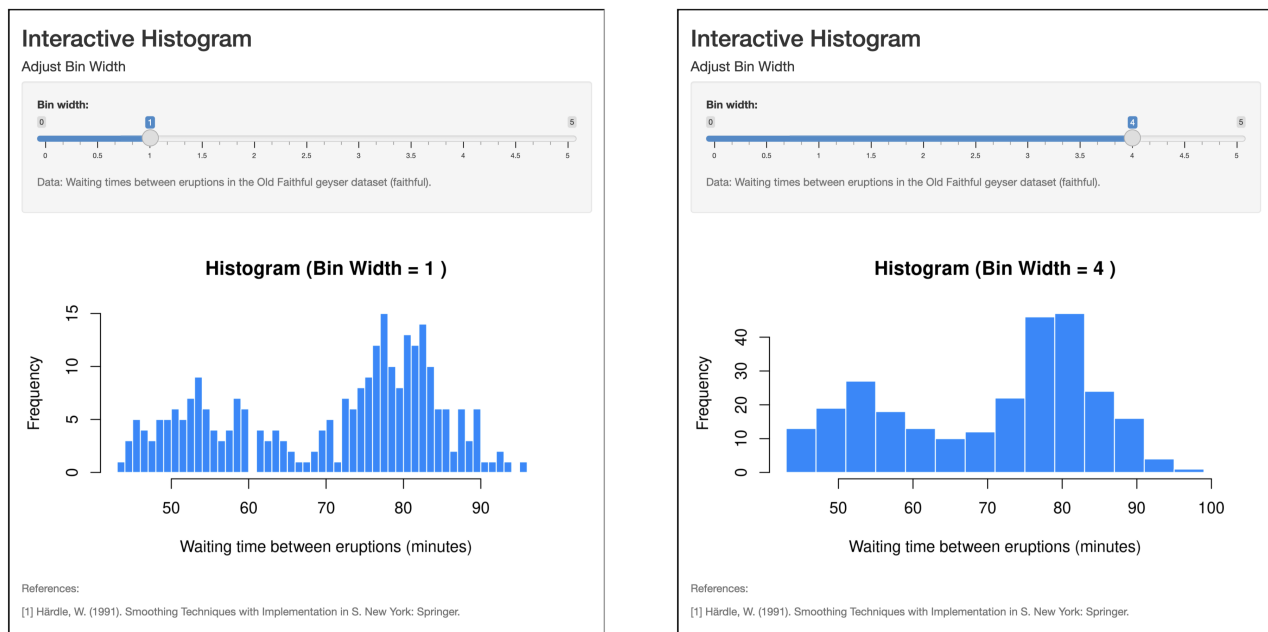
Figure 3.6: An illustration of a histogram to have a symmetric probability distribution.

(b) Interactive histogram created with `plotly` with density plot shown from Section 3: Descriptive Statistics.

Figure 5: Interactive plot created with `plotly` which will can or hide the density plot when the user clicks the *Show/Hide Density* button on the plot.

4 CONCLUSIONS, REFLECTION AND FUTURE WORK

Beyond its technical and instructional outcomes, this project demonstrates the pedagogical potential of student-led authorship and design. By actively participating in the creation of educational content, students shift from consumers to contributors of knowledge, engaging more deeply with the subject matter.



(a) Interactive histogram with the slider for bin width set to a bin width of 1 minute.

(b) Interactive histogram with the slider for bin width set to a bin width of 4 minutes.

Figure 6: Interactive application that allows students to adjust the bin width for a histogram built on real-world data. The specific app was created using the *faithful* data set which has the waiting times for the Old Faithful geyser in Yellowstone National Park. The tool demonstrates how histogram appearance is sensitive to graphical parameters, enhancing students' understanding of distribution shape and granularity.

The process required not only statistical expertise and programming fluency, but also adaptability, design thinking, and critical reflection—skills applicable in both academic and professional contexts. Through collaborative authorship, the final product embodies negotiated decisions, iterative problem-solving, and responsiveness to peer and faculty feedback.

More broadly, the project presents a replicable model of participatory curriculum development that integrates open-source tools, student initiative, and faculty mentorship to generate accessible, engaging, and adaptive learning resources.

One of the most time-consuming components was designing figures and interactive plots. Although this was challenging, it was also one of the most rewarding aspects of this project. Many were created using `ggplot2` and are now central features of the e-book. They are visually clear, carefully styled, and interactive where appropriate. These visualizations reflect intentional design choices and help communicate complex ideas in a way that is both intuitive and accessible. Although this process was time-consuming, it was also one of the most enjoyable and educational parts of the project.

While the current version of the e-book successfully integrates core statistical ideas with interactive elements, there are several directions for future development. One important area is expanding the content to include more advanced topics, such as generalized linear models, non-parametric tests, or multivariate analysis. Including these topics would make the resource more useful for students in upper-year or specialized courses.

Improving accessibility and the overall user experience is another key goal. Possible next steps include optimizing the interface for mobile devices, adding text-to-speech functionality, and offering multilingual options.

These features would help the e-book reach a wider range of learners from diverse backgrounds.

From a technical perspective, future versions could explore more extensive use of Shiny dashboards, where students would be able to interact with full datasets and generate real-time statistical outputs within the text-book. Additional features, such as embedded quizzes, automatic feedback on code, and user-submitted notes, could further enhance student engagement and retention of concepts.

In the long term, we hope to maintain and grow the project through continued collaboration between students and faculty. Building a clear versioning system, contributor documentation, and a guide for future authors would help ensure that the e-book remains a living resource that evolves with each new cohort.

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